



Intensive Care and Transport Ventilator Solutions



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LYRA x1 **Best performing and versatile ventilation for hospital applications**

A compact turbine driven ventilator with multi-function, covers the non-invasive and invasive ventilation, and is suitable for treatment of most patient type. LYRA x1 is versatile throughout hospital and transport. Comprehensive ventilating modes, including APRV, PRVC, NIV are available for all your demands and for all type of patients from neonatal to adult.

A collapsible high-resolution touch-screen display makes LYRA x1 mounted on a trolley your choice for ICU applications, as well as a high performance ventilator throughout hospital and transport.

The innovative expiration valve disassembling concept brings more ease and efficiency for the sterilization process. As your versatile assistant, LYRA x1 is configured with O2 therapy, P-V tool, a lung titrating gold standard, etc.



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LYRA x1

Technical Specifications

Physical Specification

Dimensions: 336 mm x 330 mm x 345 mm
(L x W x H): 664 mm x 600 mm x 1370 mm
(with trolley)
Weight: Approximately 9.5 kg,
Approximately 31.0 kg (with trolley)

Screen

Display Size: 12.1 Color active matrix TFT touch
Display Resolution (H) x (V): 1280 x 800 pixels
Brightness: Adjustable

Ventilation Specifications

Patient Type: Adult, Pediatric, Neonate

Invasive Ventilation Mode:

VCV (Volume Control Ventilation)

PCV (Pressure Control Ventilation)

VSIMV (Volume Synchronized Intermittent Mandatory Ventilation)

PSIMV (Pressure Synchronized Intermittent Mandatory Ventilation)

CPAP/PSV (Continuous Positive Airway Pressure/Pressure Support Ventilation)

PRVC (Pressure Regulated Volume Control)

V + SIMV (PRVC + SIMV)

BPAP (Bilevel Positive Airway Pressure)

APRV (Airway Pressure Release Ventilation) 
Apnea Ventilation

Non-invasive Ventilation Mode:

PCV (Pressure Control Ventilation)

PSIMV (Pressure Synchronized Intermittent Mandatory Ventilation)

CPAP/PSV (Continuous Positive Airway Pressure/Pressure Support Ventilation)

BPAP (Bilevel Positive Airway Pressure)

APRV (Airway Pressure Release Ventilation) 

O2 Therapy

High Flow Oxygen Treatment

Controlled Parameters

O₂ %: 21-100% (increments of 1%)
VT (Tidal Volume): Adult: 100-2000 mL
(increments of 10 mL) / Pediatric: 20-300 mL /
Neonate: 2-300 mL (increments of 1 mL)
f (Ventilation frequency): 1-80 bpm /
Neonate: 1-150 bpm (increments of 1 bpm)
fSIMV (Ventilation frequency in SIMV mode):
1-80 bpm / Neonate: 1-150 bpm
(increments of 1 bpm)
I:E range: 4:1-1:10 (increments of 0.5)
T_{insp} (Inspiratory time): 0.20-10 s
(increments of 0.05 s)
T_{slope} (Time of Pressure Rising): 0-2.00 s (in-
crements of 0.05 s)
T_{high}: 0.2-30 s (increments of 0.1 s)
T_{low}: 0.2-30 s (increments of 0.1 s)
T_{pause}: 5%-60% (increments of 1%), Off 
ΔP_{insp}: 5-60 cm H₂O (increments of 1 cm H₂O)
ΔP_{supp}: 0-60 cm H₂O
(increments of 1 cm H₂O)
P_{high}: 0-60 cm H₂O (increments of 1 cm H₂O)
P_{low}: 0-45 cm H₂O (increments of 1 cm H₂O)
PEEP: 1-45 cm H₂O
(increments of 1 cm H₂O), Off 
Flow trigger: 0.5-15 L/min
(increments of 0.1 L/min)
Pressure trigger: -10 to -0.5 cm H₂O
(increments of 0.5 cm H₂O)
Exp% (Expiration termination level): 10-85%
(increments of 5%), Auto

LYRA x1

Technical Specifications

Apnea Ventilation

V_tapnea: Adult: 100-2000 mL (increments of 10 mL) / Pediatric: 20-300 mL / Neonate: 2-300 mL (increments of 1 mL)
 Δ Papnea: 5-60 cm H₂O (increments of 1 cm H₂O)
fapnea: 1-80 bpm (increments of 1 bpm)
Apnea T_{insp}: 0.20-10 s (increments of 0.05 s)

Sigh

Sigh Switch: On, Off
Interval: 20 s-180 min (increments of 1 s from 20 to 59 s, increments of 1 min from 1 to 180 min)
Cycles Sigh: 1-20 (increments of 1)
 Δ int.PEEP: 1-45 cm H₂O (increments of 1 cm H₂O), Off

Synchronized Tube Resistance Compliance

Tube Type: ET Tube, Trach Tube, Disable STRC
Tube I.D.: Adult: 5.0 -12.0 mm (increments of 0.5 mm) / Pediatric: 2.5 - 8.0 mm (increments of 0.5 mm)
Compensate: 0-100% (increments of 1%)
Expiration Compliance Switch: On, Off

Monitored parameters

Numeric:

Paw	V _t e	C _{dyn}
P _{peak}	V _{Ti}	C _{stat}
P _{plat}	Oxygen concentration	R _{cexp}
P _{mean}	V _{Te} spn	WOB
PEEP	V _{Te} /IBW	RSBI
Insp Flow	f _{total}	NIF
Exp Flow	f _{mand}	P0.1
MV	f _{spn}	PEEPi
MV leak	Re	Continuous Flow (O ₂ Therapy)
MV spn	Ri	

Real time Graphics:

Pressure-time waveforms: Paw-Volume Loop

Flow-time waveforms: Flow-time Loop

Volume-time waveforms: Paw-Flow Loop

Control Accuracy

O₂%: $\pm(3 \text{ vol.\%} + 1\% \text{ of setting})$
TV: $\pm(10 \text{ mL} + 10\% \text{ of setting})$ (BTPS)
T_{insp}: $\pm 0.1 \text{ s}$ or $\pm 10\% \text{ of setting}$, whichever is greater
I: E 2:1 to 1:4: $\pm 10\% \text{ of setting}$, other range: $\pm 15\% \text{ of setting}$
f: $\pm 1 \text{ bpm}$
fSIMV: $\pm 1 \text{ bpm}$
T_{slope}: $\pm(0.2 \text{ s} + 20\% \text{ of setting})$
PEEP: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
 Δ P_{insp}: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
 Δ P_{supp}: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
P_{high}: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
P_{low}: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
Thigh: $\pm 0.2 \text{ s}$ or $\pm 10\% \text{ of setting}$, whichever is greater
T_{low}: $\pm 0.2 \text{ s}$ or $\pm 10\% \text{ of setting}$, whichever is greater
Pressure Trigger: $\pm(1.0 \text{ cm H}_2\text{O} + 10\% \text{ of setting})$
Flow Trigger: $\pm(1.0 \text{ L/min} + 10\% \text{ of setting})$
 Δ int.PEEP: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
Exp%: $\pm 10\%$
fapnea: $\pm 1 \text{ bpm}$
 Δ Papnea: $\pm(2.0 \text{ cm H}_2\text{O} + 5\% \text{ of setting})$
T_{vapnea}: $\pm(10 \text{ mL} + 10\% \text{ of setting})$ (BTPS)
Apnea T_{insp}: $\pm 0.1 \text{ s}$ or $\pm 10\% \text{ of setting}$, whichever is greater

Monitoring Accuracy

Airway pressure (Ppeak, Pplat, Pmean, PEEP, PAP, EPAP): $\pm(2 \text{ cm H}_2\text{O} + 4\%$ of the actual reading)

Tidal Volume: (Tvi, Tve, TVe/IBW, Tve spn):
0 ml-100 ml: $\pm(10 \text{ ml} + 3\%$ of the actual reading) (BTPS)

100 ml-4000 ml: $\pm(3 \text{ ml} + 10\%$ of the actual reading) (BTPS)

Minute Volume (MV, MVspn, Mvleak):
 $\pm 0.3 \text{ L/min}$ or $\pm 8\%$ of the actual reading, whichever is greater (BTPS)

Frequency (ftotal, fmand, fspn): $\pm 5\%$ of reading or $\pm 1 \text{ bpm}$, whichever is greater

Inspired Oxygen (FiO₂): $\pm(2.5 \text{ vol.}\% + 2.5\%$ of the actual reading)

Resistance: 0 to 50: $\pm 10 \text{ cm H}_2\text{O/L/s}$

Other range: 50% of the actual reading

Compliance: 25% of the actual reading or $\pm 10 \text{ ml/cm H}_2\text{O}$, whichever is greater

RSBI: 0 to 999 1/(min*L): $\pm (3 \text{ 1/(min*L)} \pm 15\%$ of the actual reading)

WOB: -

NIF: $\pm(2 \text{ cm H}_2\text{O} + 4\%$ of the actual reading)

P0.1: $\pm(2 \text{ cm H}_2\text{O} + 4\%$ of the actual reading)

PEEPi: -

Rcexp: -

Alarm settings

Tidal Volume: High / Low

Minute Volume: High / Low

Airway pressure: High / Low

Frequency: High / Low

Inspired Oxygen (FiO₂): High / Low

etCO₂: High / Low

Apnea alarm time: 5-60 s

Trend

Type: Tabular, Graphic

Length: 72 hours

Content: Monitor Parameters, Setting Parameters (Setting Ventilation mode and Parameters)



LYRA x1

Technical Specifications

Controlled Parameters

O₂ %: 21-100% (increments of 1%)

Flow: 4-60 L/min

Controlled Accuracy

O₂ %: $\pm(3 \text{ vol.}\% + 1\% \text{ of setting})$

Flow: $\pm(2 \text{ L/min} + 10\% \text{ of setting})$ (BTPS)

Environmental specifications

Temperature: 5-40 °C (operating); -20 to 60 °C (storage and transport, O₂ sensor: -20 to 50 °C)

Relative Humidity: 10-95% (operating);

10-95% (storage and transport)

Barometric Pressure: 62-106 kPa (operating);

50-106 kPa (storage and transport)

Power Battery Backup

External AC power supply

Input voltage: 100-240 V

Input frequency: 50/60 Hz

Input current: 2.5 A Max

Fuse: T2.5 AH/250 V

Internal battery

Number of batteries: One or Two (Optional)

Battery type: Build-in Lithium-ion battery, 11.25 VDC, 6400 mAh

Battery run time: 3 hours (Powered by one new fully-charged battery in standard working condition), 6 hours (powered by two new fully-charged batteries in standard working condition).

Others

Communication interface: RS-232, Ethernet,

USB port, CO₂ analyzer connector

Gas supply: O₂

(HPO) Oxygen connector: NIST (DISS optional)

Gas supply pressure: 280-600 kPa





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For more information, please contact us.

aXcent medical GmbH

Josef-Görres-Platz 2
56068 Koblenz
Germany

Phone: +49 261 30 11 117

Fax: +49 261 30 11 111

Mail: info@axcentmedical.com



www.axcentmedical.com

Alkaram - Medical

📍 10th of Ramadan City, Alordonia, Al Jawhara Mall, Office 447.

📍 32 Obur Bulding - Salah Salem Street, Cairo, Egypt.

✉ sales@alkaram-medical.com

🌐 www.alkaram-medical.com

☎ 01099442054 - 01206467528

☎ 01272398884 - 01032671638



www.alkaram-medical.com